Saliency-Guided Single Shot Multibox Detector for Target Detection in SAR Images

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***Abstract*— The single shot multibox detector (SSD), a proposal-free method based on convolutional neural network (CNN), has recently been proposed for target detection and has found applications in synthetic aperture radar (SAR) images. Moreover, the saliency information reflected in the saliency map can highlight the target of interest while suppressing clutter, which is beneficial for better scene understanding. Therefore, in this article, we propose a saliency-guided SSD (S-SSD) for target detection in SAR images, in which we effectively integrate the saliency into the SSD network not only to suggest where to focus on but also to improve the representation capability in complex scenes. The proposed S-SSD contains two separated convolutional backbone subnetwork architectures, one with the original SAR image as input to extract features, and the other with the corresponding saliency map obtained from the modified Itti's method as input to acquire refined saliency information under supervision. In addition, the dense connection structure, instead of the plain structure used in original SSD, is applied in the two convolutional backbone architectures to utilize multiscale information with fewer parameters. Then, for integrating saliency information to guide the network to emphasize informative regions, multilevel fusion modules are utilized to merge the two streams into a unified framework, thereby making the whole network end-to-end jointly trained. Finally, the convolutional predictors are used to predict targets. The experimental results on the miniSAR real data demonstrate that the proposed S-SSD can achieve better detection performance than state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), deep learning, saliency detection, single shot multibox detector (SSD), synthetic aperture radar (SAR), target detection.**

I. Introduction

OWING to the well-known fact that the synthetic aperture radar (SAR) can provide remote sensing images under any conditions of the day and the weather, the SAR imaging technology has gained rapid development in recent years. With the overwhelming amount of SAR images available, SAR automatic target recognition (ATR) develops rapidly. The most popular SAR ATR system consists of three stages: target detection [1]–[4], target discrimination [5], [6], and target recognition [7], [8].

Manuscript received May 31, 2019; revised October 15, 2019; accepted October 25, 2019. Date of publication December 11, 2019; date of current version April 22, 2020. This work was supported in part by the National Science Foundation of China under Grant 61771362, Grant U1833203, and Grant 61671354, in part by the 111 Project under Grant B18039, and in part by the Shaanxi Innovation Team Project. *(Corresponding author: Lan Du.)*

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Digital Object Identifier 10.1109/TGRS.2019.2953936

The demand for SAR target detection, the first stage in the SAR ATR, is increasing sharply. Among various target detection methods available for SAR images, the constant false alarm rate (CFAR) methods [1], [2] have raised lots of concerns and are extensively studied. Resorting to the clutter statistical distribution model, the CFAR methods use the given false alarm rate and the clutter statistical characteristics to compute the adaptive threshold and then compare the intensity of the pixel to be detected and the threshold to determine whether the pixel is a target or not. By regarding the Gaussian distribution as the background clutter statistical distribution model, the two-parameter CFAR [1] acquires good performance in some simple SAR images. A Gamma-based CFAR detector proposed by Alberola-Lopez *et al.* [2] assumes that the background clutter obeys the Gamma statistical distribution. The CFAR methods achieve good performance in simple scenes. However, when the scene is complex in practice, such as urban areas, an appropriate clutter statistical distribution model is difficult to select and the performance will be degraded when the clutter statistical distribution model is unsuitable [9]. For example, in the complex ground scene of the miniSAR data which are used for the experiments, the targets to be detected are vehicles and the clutter includes man-made buildings, roads, trees, grasslands and so on, the CFAR methods cannot gain a preferable performance, only less than 40% in precision and about 80% in recall. A satisfactory performance that we expect is more than 0.9, both in precision and recall.

The convolutional neural networks (CNNs) have achieved significant success and have become the dominant machine-learning approach. Being data driven, the CNNs can automatically learn the extraction of features to gain better performance in the field of computer vision in the past few decades. In the field of target recognition, based on CNN there are many pieces of research. Micro CNN [10], which is small, fast and accurate, is designed for real-time recognition. Reference [11] introduces a noise-invariant regularization into the CNN for reducing the influence of noise. Cui *et al.* [12] propose a semi-supervised target recognition method based on the CNN and an assistant support vector machine (SVM) decision. Also, the CNN is used for the digital elevation estimation [13], semantic segmentation, adversarial attacks, and defense, etc. In the field of target detection, there are many deep detection networks that have been proposed. These detection networks at first need to be trained by the training images and corresponding labels and then take the original image as input to generate the detection results when testing. Actually, the deep learning-based detection network methods deliver good performance and they do not need the clutter statistical modeling. Generally, these target detection frameworks fall into two groups. One is the two-stage detectors. The region-based CNN (R-CNN) method [14] is firstly proposed. Then, based on R-CNN, the Fast R-CNN [15] and the Faster R-CNN [16] are proposed with higher speed and accuracy. The other is the one-stage detectors such as you only look once (YOLO) [17] and single shot multibox detector (SSD) [18]. Based on an end-to-end deep neural network without the region proposal extraction step, YOLO and SSD are simpler and faster relative to the methods that require object proposals. Compared with YOLO, SSD can further improve the detection accuracy via multilevel feature prediction and the application of anchor. Besides target detection in optical images, the SSD also achieves good performance in SAR images. Wang *et al.* [19] applied the SSD into the target detection for SAR images and this proposed method achieved better performance than the conventional SAR target detection methods.

However, the scene of SAR images is complex, especially the background clutter which contains buildings, roads, grasslands, trees, streetlights, and so on. The complex background clutter hampers the performance of the SAR target detection, making it difficult to have fewer missing alarms and false alarms. In recent years, saliency detection, which imitates the ability of the human visual attention system has received a great deal of attention. Aiming at highlighting certain regions or objects to which one's attention can be attracted in a scene and suppressing the background clutter, saliency detection can allocate more computational resources for better scene understanding. Lots of image saliency detection methods have been proposed. Focusing on the center-surround difference, the Itti's method [20], a classical and well-used saliency detection method, extracts multiscale feature maps using dyadic Gaussian pyramids, and then the difference between central region and its surrounding region is computed by point-to-point subtraction on two feature maps as the measurement of saliency to generate saliency map. Liu *et al.* [21] proposed a visual attention detection method that combines Itti's pyramid model with the singular value decomposition (SVD). Considering the prior knowledge of targets to be detected, Wang *et al.* [22] modified Itti's saliency model and achieved better performance by introducing the target's prior size information to select some task-dependent scales in Gaussian pyramids for SAR target detection.

According to the above analysis, in this article, a novel saliency-guided SSD (S-SSD) for target detection in SAR images is proposed, in which the saliency information is integrated into the SSD network to guide the network to emphasize informative regions. More specifically, the main contributions of the proposed target detection method are as follows.

1) The saliency information is introduced into a deep learning-based target detection network. Through draw support from the characteristic of saliency information, i.e., highlighting targets and suppressing clutter, the deep learning-based target detection network can be guided to know where to focus on more intelligent and has a better scene understanding capability. With the saliency maps obtained by the modified Itti's method as inputs, one convolutional backbone architecture is used to learn the refined saliency information under supervision. The refined saliency information is beneficial for emphasizing the informative regions and improving the representation capability, and thereby a significant improvement in detection performance is achieved.
2) The dense connection structure, instead of the plain structure in the original SSD, is used in the proposed S-SSD. The dense connection structure integrates lower-level features and higher-level features, which can introduce more context information and mine the correlated complementary advantages across scales with fewer parameters. In the dense connection structure, half of the feature maps are learned from the previous feature maps, which is more abstract. The other half is the reuse of the previous features, where features have more detailed information. By integrating the above-mentioned lower-level feature and higher-lever feature, the multiscale feature can be used comprehensively and effectively.
3) For the integration process of the saliency information and the deep learning-based target detection network, a fusion module is proposed in this article, which can explicitly utilize the characteristics of saliency information. Different from the original concatenation module, the proposed fusion module element-wise multiply the saliency map with the feature maps, which can merge the saliency information into feature maps with fewer parameters and achieve better performance.

The experiments on the miniSAR real data demonstrate that the proposed target detection method outperforms than the conventional SAR target detection method in terms of detection performance.

The remainder of this article is organized as follows. Section II introduces the proposed novel target detection method. Section III gives the experimental results and analysis based on the miniSAR real data. Finally, Section IV concludes this article.

II. S-SSD FOR SAR TARGET DETECTION

Fig. 1 shows the flowchart of our proposed S-SSD architecture for target detection in SAR images.

As shown in Fig. 1, the proposed S-SSD contains two separated convolutional backbone sub-network architectures. Each backbone sub-network is combined with the truncated VGG16 network and some added convolutional layers. In the two streams, one takes the original SAR images as input. This stream effectively extracts multiscale feature maps with different resolutions. The other takes their corresponding saliency map, obtained from the modified Itti's method [22], as input. Under supervision, this stream can acquire more refined saliency information at corresponding scales. In addition, the dense connection structure is adopted in the added convolution layers for utilizing multiscale information with fewer parameters. Then, the above-mentioned two streams

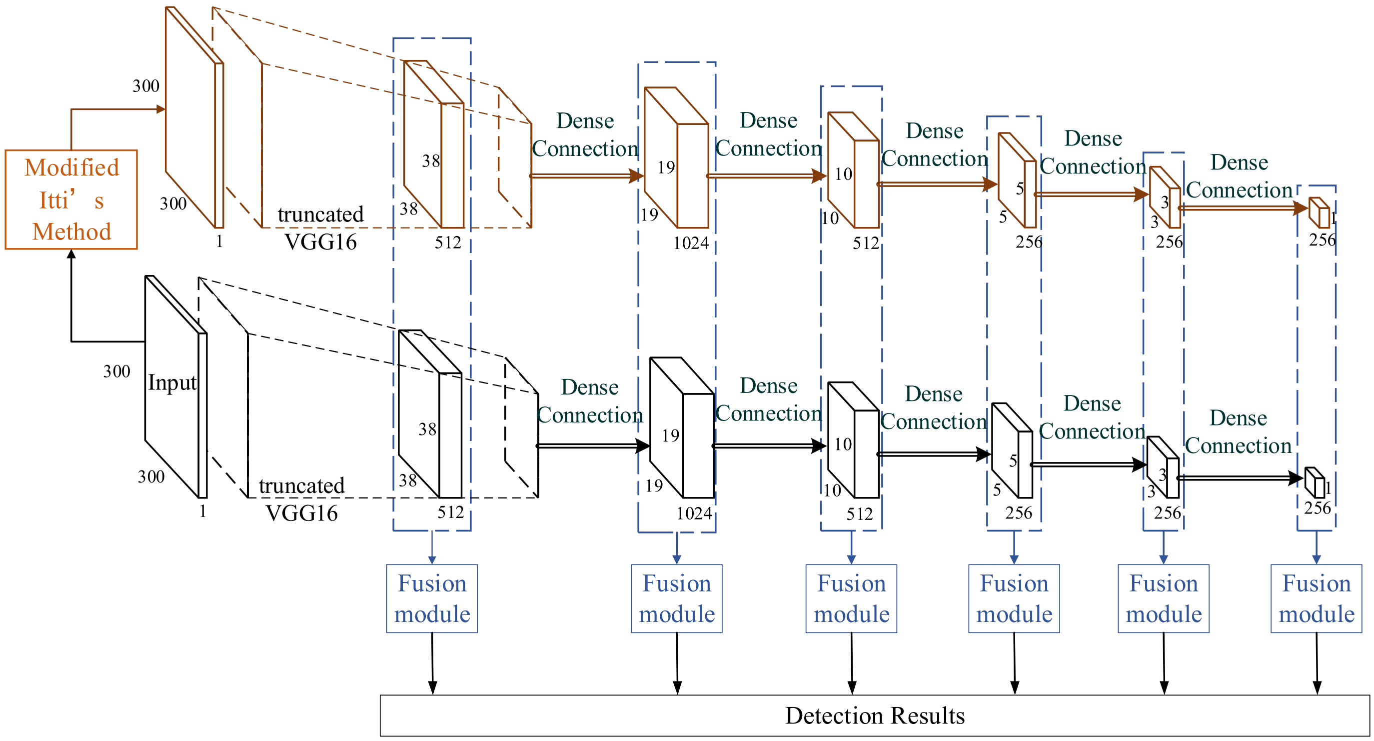


Fig. 1. Flowchart of the proposed S-SSD target detection method in SAR images.

are merged into a unified framework by multiple fusion modules at multiple levels. Via the fusion module, the saliency information is integrated to guide the network to emphasize informative regions. Finally, these multilevel fused feature maps are fed into convolutional predictors to predict targets and their offsets in bounding box locations.

In Sections II-A–II-D, some important components, including the modified Itti's method, convolutional backbone sub-network, dense connection structure, fusion module, are introduced concretely.

A. Modified Itti's Method

As a biologically inspired visual attention model, Itti's method [20] focuses on the center-surround difference to generate the saliency map. Although the Itti's method is well used and has simple calculations, it does not consider the detection task. Therefore, based on the prior size information of the targets to be detected, the modified Itti's method is proposed.

The modified Itti's method, first, extracts multiscale feature maps using progressively low-pass filtering and subsampling on the SAR image, called dyadic Gaussian pyramids. Second, based on the prior size information of the detection task, the appropriate surround scales s and the appropriate center scale c are selected. Appropriate surround scales can make for excluding some large clutter and center scale is used to exclude some small clutter. Then, a center-surround operation, which is implemented via the across-scale difference between the center pixel at the selected center scale c in the Gaussian pyramids and the corresponding surround pixel at the selected surround scales, is used to obtain multiple feature maps. Finally, the saliency map is generated by linearly fusing these feature maps.

B. Backbone Sub-Networks

The two backbone sub-networks in our proposed S-SSD are the same and each one is combined with the truncated VGG16-based network and some added convolutional layers. These layers decrease in size (i.e., increase in resolution) progressively. By using multiple layers with different resolutions, the S-SSD can perceive targets of various sizes.

The base network adopts the truncated VGG16 network. With a stack of convolutional layers, where the convolution filters are very small (only 3×3), the VGG networks have deeper architectures to achieve better representation. It is easy to see that, a stack of two convolutional layers with 3×3 convolution filters has the same effective receptive field with one convolutional layer with 5×5 convolution filters. However, the stack of two convolutional layers with 3×3 convolution filters increases the nonlinearity and decreases the number of parameters compared with the single convolutional layers with 5×5 convolution filters.

Followed by the truncated VGG16-based network, some convolutional layers are added to further extract feature maps with different resolutions, which is also called auxiliary structure. In the added convolution layers, the dense connection structure, instead of the plain structure used in original SSD, is adopted. The introduction of the dense connection structure can utilize multiscale context information and the dense connection structure has fewer parameters than the original plain structure. The concrete illumination of the dense connection structure is described in Section II-C.

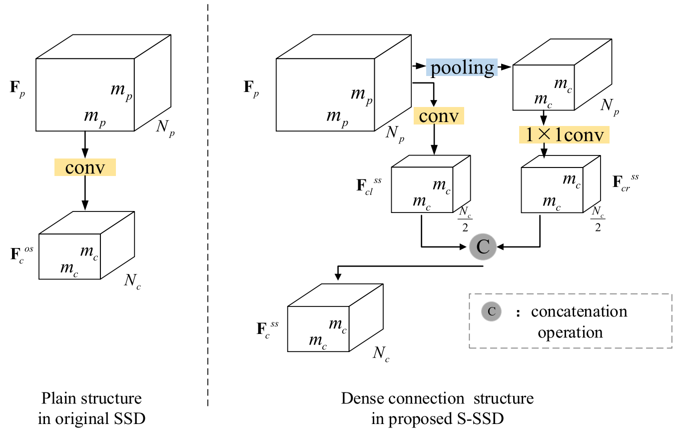


Fig. 2. Flowchart of the plain structure in original SSD and the dense connection structure in the proposed S-SSD.

C. Dense Connection Structure

The original SSD extracts multiple feature maps with different resolutions and via using multiple layers to predict targets, the SSD achieves the goal that its detection network can detect targets of various sizes. However, on a single scale, the SSD only utilizes the current feature maps to predict targets. It is well known that the lower-level features contain more detail information, and the higher-level features care more about semantic information. Fusing multiscale features can introduce more context information and mine the correlated complementary advantages across scales, which is effective for feature representation. Therefore, we adopt the dense connection to fuse multiscale information before prediction.

Fig. 2 shows the original plain structure in SSD and the dense connection structure in the proposed S-SSD. From Fig. 2 we can see that, in the original SSD, the current feature maps $\mathbf{F}_c^{\text{os}} \in \mathbb{R}^{m_c \times m_c \times N_c}$ are directly learned from the previous feature maps $\mathbf{F}_p \in \mathbb{R}^{m_p \times m_p \times N_p}$ the resolutions of which are higher than $\mathbf{F}_c^{\text{os}}$, where m_c and N_c are the spatial size and the channel dimension of $\mathbf{F}_c^{\text{os}}$, m_p and N_p are the spatial size and the channel dimension of $\mathbf{F}_p$. Nevertheless, in the dense connection structure, on the one hand, we use 3×3 convolution layers (followed by a max pooling layer) to learn higher-level feature maps $\mathbf{F}_{\text{cl}}^{\text{ss}} \in \mathbb{R}^{m_c \times m_c \times (N_c/2)}$, which has only half of N_c channels. On the other hand, at first, we utilize the max pooling layer to down-sample the feature maps $\mathbf{F}_p$. The spatial size of the down-sampled feature maps is also m_c. Then, a convolution layer with 1×1 filters is appended to reduce the channel dimension from N_c to $(N_c/2)$. The generated feature maps $\mathbf{F}_{\text{cr}}^{\text{ss}} \in \mathbb{R}^{m_c \times m_c \times (N_c/2)}$ reuse the information of $\mathbf{F}_p$. Finally, the feature maps $\mathbf{F}_{\text{cl}}^{\text{ss}}$ and $\mathbf{F}_{\text{cr}}^{\text{ss}}$ are concatenated as the current feature maps $\mathbf{F}_c^{\text{ss}} \in \mathbb{R}^{m_c \times m_c \times N_c}$.

Although $\mathbf{F}_c^{\text{os}}$ and $\mathbf{F}_c^{\text{ss}}$ have the same size and the number of channels, $\mathbf{F}_c^{\text{ss}}$ fuses multiscale information which is useful for subsequent prediction. Because half of feature maps in $\mathbf{F}_c^{\text{ss}}$ come from the reuse of previous feature maps, only the other half need to learn, the dense connection structure needs fewer parameters than the plain structure.

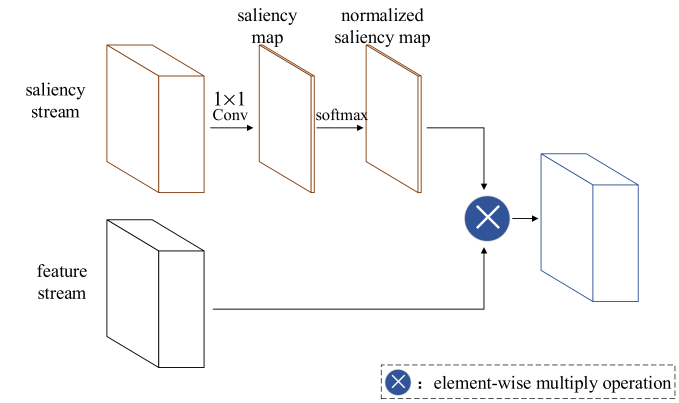


Fig. 3. Flowchart of the fusion module.

D. Fusion Module

In this article, the S-SSD contains two streams, one of which extracts features and the other acquires corresponding saliency information. To take advantage of the saliency information which can highlight targets of interest and suppress clutter, we develop a fusion module to integrate the saliency information into the feature maps.

Fig. 3 shows the fusion module. At first, we utilize 1×1 convolution filters, which can be seen as a combination of the input channels in saliency stream, to generate the saliency map. The value s_i of a spatial position i in the saliency map represents the saliency score. The higher the saliency score is, the more likely the spatial position (or corresponding receptive field) is target. Then, the saliency map is normalized by a softmax operation

$$\text{ns}_i = \frac{\exp(s_i)}{\sum_{j=1}^{n} \exp(s_j)}, \quad i \in \{1, 2, \ldots, n\} \tag{1}$$

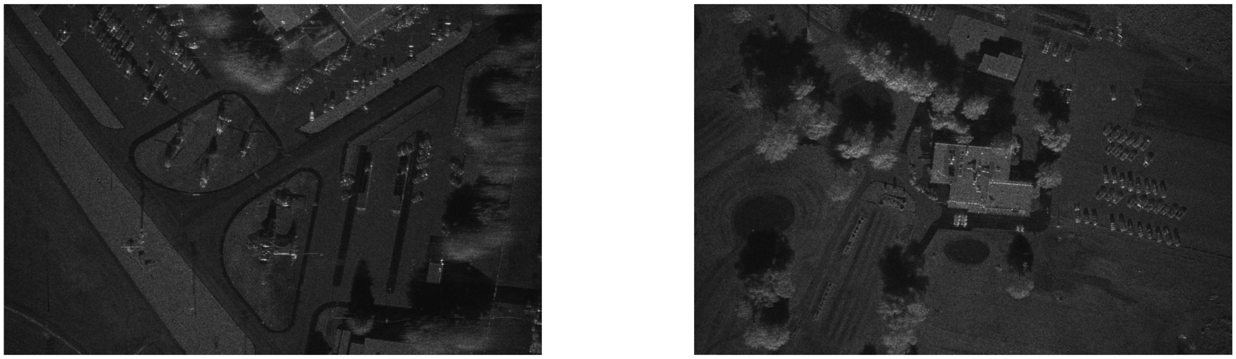
where ns_i represents the normalized saliency score of spatial position i, n represents the total number of the spatial position in saliency map, $\exp(\cdot)$ represents the standard exponential function. Finally, the normalized saliency map is element-wise multiplied to each input feature map in feature stream to generate the final feature maps.

By the fusion module, the saliency information is integrated into the feature maps to improve the representation power. Because the proposed S-SSD also utilizes multiple feature maps with different resolutions to predict targets, multiple saliency map are learned at the corresponding levels. Thus multiple fusion modules are used in abovementioned levels.

III. Experimental Results and Analysis

A. Experimental Data Description

The measured miniSAR data set is acquired by U.S. Sandia National Laboratories in 2005. Nine SAR images in the miniSAR data set are selected for our experiments. The resolutions of these SAR images are 0.1 m $\times$ 0.1 m and the sizes are 1638 $\times$ 2510 pixels. Among the nine SAR images, seven images are used for training and the rest for testing. Here, we give the test SAR images in Fig. 4. From Fig. 4 we can



(a) (b)

Fig. 4. Two original test SAR images. (a) Original image I. (b) Original image II.

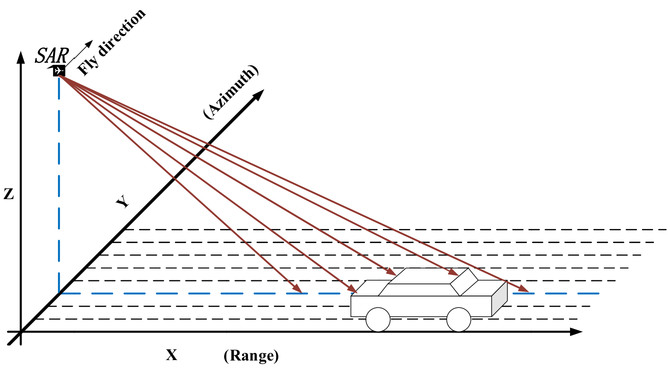


Fig. 5. The geometry of the acquisition in the SAR scene space.

TABLE I
CHARACTERISTICS OF THE MINISAR DATA ACQUIRED ON 19TH MAY 2005

System Characteristics	Numerical Value
Mode	Spotlight
Band	Ku
Range resolution	4-inch
Azimuth resolution	4-inch
Swath Middle Range	3.3km
Incidence Angle	26°-29°
Center Frequency	16.8GHz

see that the SAR images contain many vehicle targets and the background of SAR images is complex, including buildings, roads, grasslands, trees, and streetlights. For convenience, we refer to the two SAR images as image I and image II, respectively. The simplified geometry of the acquisition in the SAR scene space is shown in Fig. 5, with the X-axis in the range direction, Y-axis in the azimuth direction, and Z-axis in the height direction; and in the scenes, the vehicles are the targets to be detected. Table I gives the acquisition parameters of the SAR sensor [23]–[25].

B. Training Settings

For the parameter initialization of the truncated VGG16-based network, we transfer the VGG16 model pre-trained on the MSTAR data set to the truncated VGG16 at the corresponding layers. For other parameters in the S-SSD, we initialize the parameters in convolutional layers with the Xavier method. Under the weighted loss function, i.e., smooth L1 loss for localization and softmax loss for classification, the proposed network is jointly optimized, i.e., backpropagation is applied end-to-end using stochastic gradient descent (SGD) optimizer. For the miniSAR data, the center scale in the modified Itti's method for generating saliency map is set to $c = \{2\}$ and the surround scales are set to $s = \{4, 5, 6\}$ [22].

Due to the size restriction of the input image of general networks, in a practice context, the large SAR image needs to be cropped into several adjacent image blocks for training and testing. For the proposed S-SSD methods, we crop the SAR image into several image blocks the size of which is 300×300 pixels according to the size of the input layer. When training, the image blocks of train data are used to train the network. When testing, each image block is processed individually. Thereafter, according to the order of the division, the adjacent image block's detection results are stitched together. And the experiments are implemented with the Caffe deep-learning framework, using a personal computer with Intel Xeon E5-2620v4 CPU of 2.10 GHz and NVIDIA GTX 2080Ti GPU on a Ubuntu 16.04 Linux system.

C. Evaluation Criteria

We quantitatively evaluate different detection methods via four widely used criteria, namely precision, recall, $F1$-score and the figure of merit (FoM). The precision measures the fraction of detections that are true positives. It is formulated as

$$\text{precision} = \frac{\text{the number of target chips}}{\text{the total number of chips}}. \quad (2)$$

The recall measures the fraction of positives over the number of ground truths.

$$\text{recall} = \frac{\text{the number of detected targets}}{\text{the total number of targets}}. \quad (3)$$

And the $F1$-score, a weighted average of precision and recall, can comprehensively evaluate the quality of a detection method.

$$F1-\text{score}=\frac{2\times\text{precision}\times\text{recall}}{\text{precision}+\text{recall}}. \tag{4}$$

The definition of the FoM is as follows:

$$\text{FoM}=\frac{\text{the number of detected targets}}{\text{the number of targets}+\text{the number of false alarms}}. \tag{5}$$

The higher the values of recall, precision, $F1$-score, and the FoM are, the better the performance of target detection method is, and vice versa.

D. Comparisons With Conventional Detection Methods

In this section, we compare the proposed S-SSD method with conventional detection methods for SAR images. The conventional detection methods for comparison contain the Gaussian-CFAR (i.e., two parameters CFAR) [1], the Gamma-CFAR [2], the MI-Gaussian CFAR method, the MI-Gamma CFAR method, and the DDSR method [9]. The MI-Gaussian CFAR and the MI-Gamma CFAR are the methods in which the modified Itti's method is combined with the Gaussian CFAR and the Gamma CFAR, respectively. For combining the proposed saliency method and the CFAR method, firstly, we obtain the corresponding the saliency map via the modified Itti's method on the original SAR image. Then the saliency map is element-wise multiplied to the original SAR image to generate the fused image. Finally, based on the fused image, the CFAR method is used to generate the detection method. Fig. 6 exhibits the intuitional target detection results of the proposed method and three conventional detection methods on image I and image II, respectively. In the experimental results, green rectangles represent the correctly detected target chips, red rectangles represent false alarms and yellow rectangles represent the missing alarms. As we all know, the results of CFAR methods depend on the detection threshold. These exhibited detection methods in the following are the best results we can get through setting different thresholds.

For the miniSAR data, the ground scene is complex enough. In the SAR images, the targets to be detected are vehicles and the clutter includes man-made buildings, roads, trees, and grasslands. The CFAR methods are difficult to select a suitable clutter statistical model to gain a satisfying performance. As shown in Fig. 6, the Gaussian CFAR, the Gamma CFAR, the MI-Gaussian CFAR, the MI-Gamma CFAR, and the DDSR method has a lot of false alarms and cannot locate targets accurately. However, without clutter statistical modeling, the proposed target detection method, i.e., S-SSD, has fewer false alarms and missing alarms and achieves a more accurate location compared with the abovementioned methods under the supervision of the classification and bounding box regression.

For quantitative analysis, we count the correctly detected target chips, the missing alarms, and the false alarms of the proposed method and compared with the SAR target detection method to verify the performance. Furthermore, precision, recall, $F1$-score, and the FoM are used as evaluation criteria to quantitatively analyze the overall target detection results on the two test SAR images in Table II. Table II shows that the proposed method, S-SSD, can correctly detect more targets compared with other detection methods. Meanwhile, the missing alarms of the proposed method and the false alarms are both the fewest. From quantitative evaluation criteria, we can see that the S-SSD has the highest precision, recall, $F1$-score, and FoM. On the two test SAR images, the proposed method is about 40% higher in terms of precision, at least 7.5% higher in terms of recall, at least 28% in terms of $F1$-score and at least 42% higher in terms of the FoM than the other compared conventional SAR target detection methods. By contrast, the performance of the proposed S-SSD is superior to the other four conventional detection methods. Of course, unlike CFAR methods that are unsupervised, the proposed method is purely data-driven and becomes a drawback that cannot be ignored for most of the supervised detection methods. When the test image is totally different from the training data, the performance will be degraded and now the network needs to be retrained according to the new training data.

E. Comparisons With Deep Detection Methods

To demonstrate the superior performance of the proposed detection method, we compare against the Faster R-CNN, the original SSD, and the DA-TL SSD [19]. The Faster R-CNN well used in optical images is also introduced into SAR images to detect targets recently. The original SSD here for comparison is trained with random initialization. And the DA-TL SSD applies subaperture decomposition to acquire three-channel subaperture SAR images and then transfers the three-channel pre-train VGG model which is trained on the ImageNet to initialize the corresponding parameters. Fig. 7 gives the detection results on two SAR images.

We can observe from Fig. 7 that the proposed method has fewest missing alarms, and false alarms are compared with other deep detection methods, i.e., the Faster R-CNN, the original SSD, and the DA-TL SSD. And for quantitative analysis, Table III lists the numerical detection results on the image I and image II together. Table III shows that the proposed method, S-SSD, has the most correctly detected target chips than other deep learning-based target detection methods. Meanwhile, the amounts of the missing alarms of the proposed method are the fewest, so are the false alarms. From quantitative evaluation criteria in Table III, we can see that the S-SSD has the highest precision, recall, $F1$-score, and the FoM on the test SAR images together. Therefore, it can be concluded that the S-SSD is superior to the other abovementioned deep learning-based target detection methods.

F. Model Analyses

1) Experimental Analyses on Saliency Map: In this section, we analyze the saliency maps obtained by the modified Itti's method and our proposed method.

Fig. 8 lists the original SAR sub-images, the corresponding saliency maps via the modified Itti's method, and the corresponding saliency maps via the proposed method.

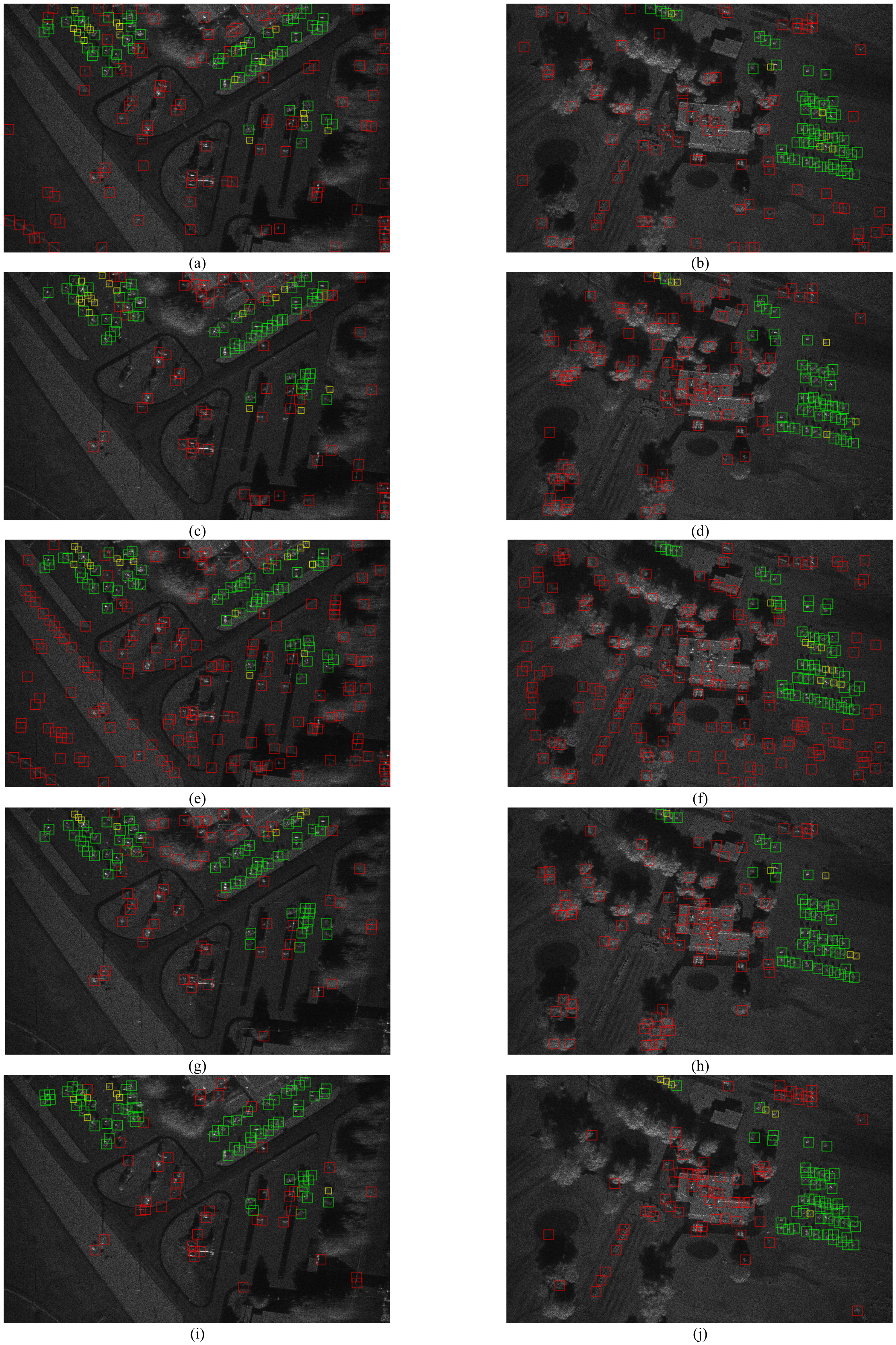
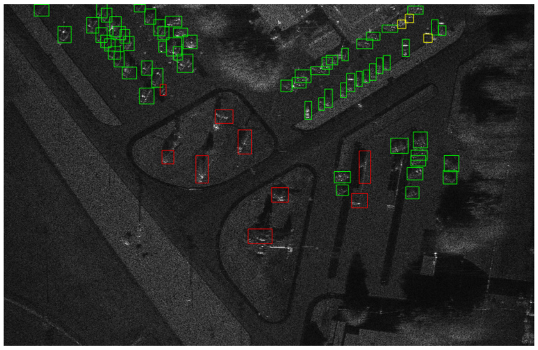


Fig. 6. Target detection results compared with conventional detection methods for images I and II, where green rectangles represent the detected target chips, red rectangles represent false alarms and yellow rectangles represent the missing alarms. (a) and (b) Gaussian-CFAR method. (c) and (d) Gamma-CFAR method. (e) and (f) MI-Gaussian CFAR. (g) and (h) MI-Gamma CFAR method. (i) and (j) DDSR method.

The modified Itti's method is an unsupervised method that is based on the center-surround difference, and it takes the prior size information of the SAR targets into consideration. Although it has a better performance on SAR images, it still



(k)

(l)

Fig. 6. *(Conituned.)* Target detection results compared with conventional detection methods for images I and II, where green rectangles represent the detected target chips, red rectangles represent false alarms and yellow rectangles represent the missing alarms. (k) and (l) Proposed S-SSD method.

TABLE II

OVERALL EVALUATION OF DIFFERENT TARGET DETECTION METHODS

	Target amounts	Detected target amounts	Missing alarms	False alarms	Chip amounts	precision	recall	F1-score	FoM
Gaussian CFAR	118	94	24	177	285	0.3789	0.7966	0.5135	0.3186
Gamma CFAR		96	22	159	262	0.3931	0.8136	0.5301	0.3466
MI-Gaussian CFAR		94	24	322	428	0.2477	0.7966	0.3779	0.2136
MI-Gamma CFAR		100	18	141	245	0.4245	0.8474	0.5656	0.3861
DDSR		105	13	107	226	0.5265	0.8898	0.6616	0.4667
S-SSD		**114**	**4**	**10**	**124**	**0.9193**	**0.9661**	**0.9421**	**0.8906**

highlights some clutter while highlighting targets. Nevertheless, in the proposed method, one of the two streams takes the saliency map obtained from the modified Itti's method as input and then some convolutional layers are used to further learn saliency information under supervision which is used in the subsequent fusion modules to improve the representation power. Fig. 7 shows that the saliency maps obtained from the proposed method highlight the targets while further suppressing clutter which cannot be suppressed via the modified Itti's method.

2) Ablation Study: To investigate each component of the proposed S-SSD method, several controlled experiments are designed on miniSAR images. From the controlled experiments, we can better understand how each component affects the performance. In all the experiments, a consistent setting is imposed unless when some components are examined. The results are summarized in Table IV.

In Table IV, the concat denotes the concatenation module and the prod denotes the proposed fusion module adopted in the S-SSD methods. From Table IV, it can be observed that the introduction of saliency information is beneficial when comparing columns 3, 5, and 6 (or columns 4, 7, and 8). Results show that the introduction of saliency information brings at last 1.5% $F1$-score improvement, which can decrease at most three missing alarms, five false alarms at the same time. And as shown in columns 5 and 6 (or columns 7 and 8), we can see that, on the basis of the introduction of saliency information, the performance improvement brought by different fusion modules is different. The concatenation module is always adopted as a fusion module in multistream network architectures [26], [27]. Nevertheless, the performance improvement of the proposed fusion module is larger than that of the concatenation. In terms of $F1$-score, the proposed fusion module is about 1% higher than the concatenation module with about three alarms. As for the dense connection structure, we can compare columns 3 and 4 (or columns 5 and 7, or columns 6 and 8) in Table IV. We observe that the case with a dense connection structure yields at least 1.3% higher $F1$-score with about three alarms than the case with a plain structure.

3) Parameters and Runtime Analysis: Compared with the original SSD network, the proposed S-SSD can run at 50 frames/s (FPS) which is lower than the speed of the SSD, while producing markedly superior detection accuracy, i.e., improving $F1$-score from 0.8638 to 0.9421 and FoM from 0.6737 to 0.8906. In terms of the parameters, the S-SSD has 45.0M parameters and the SSD only has 26.3M. It is mainly because there is only one convolutional backbone subnetwork architecture in the SSD network, which takes the image as input; whereas in the proposed method, S-SSD, due to the introduction of the saliency information, there are two separated convolutional backbone sub-network architectures,

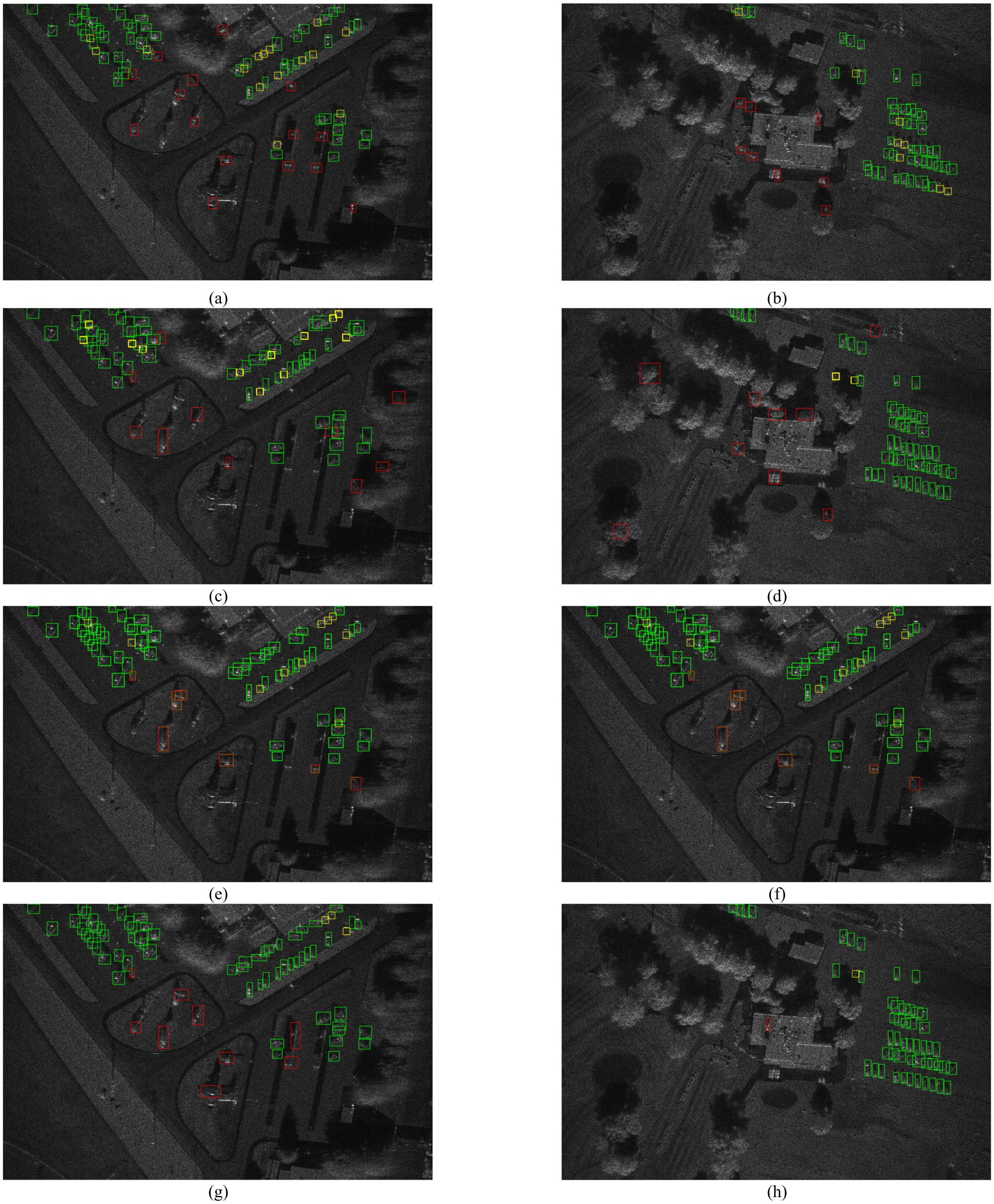


Fig. 7. Target detection results comparing with deep detection method for images I and II, where green rectangles represent the detected target chips, red rectangles represent false alarms and yellow rectangles represent the missing alarms. (a) and (b) Faster R-CNN method. (c) and (d) Original SSD method. (e) and (f) DA-TL SSD method. (g) and (h) Proposed method, i.e., S-SSD.

in which the image and its corresponding saliency map are taken as input, respectively.

Although the two backbone architecture leads to an increase in parameters, we use the dense connection structure, illustrated in Section II-C, in the two backbone architecture. On the one hand, the dense connection structure can integrate multiscale information and, on the other hand, it can reduce the parameters. Therefore, the parameters of the S-SSD are less than twice those of the SSD. And compared with the Faster R-CNN, we can see from Table V that the parameters of the S-SSD is much less than those of the Faster R-CNN, and the speed is much faster and, at the same time, the detection

TABLE III
OVERALL EVALUATION OF DIFFERENT DEEP LEARNING-BASED TARGET DETECTION METHODS

	Target amounts	Detected target amounts	Missing alarms	False alarms	Chip amounts	precision	recall	F1-score	FoM
Faster R-CNN	118	95	23	23	122	0.8115	0.8051	0.8083	0.6737
Original SSD		104	14	19	124	0.8468	0.8814	0.8638	0.7591
DA-TL SSD		106	12	14	121	0.8843	0.8983	0.8912	0.8030
S-SSD		114	4	10	124	**0.9193**	**0.9661**	**0.9421**	**0.8906**

TABLE IV
ABLATION STUDY

		S-SSD					
Components?	*Saliency stream*	×	×	✓	✓	✓	✓
	Fusion module	×	×	concat	prod	concat	**prod**
	Dense structure	×	✓	×	×	✓	✓
Quantitative evaluation	*Target amounts*	118					
	Missing alarms	9	7	7	7	5	**4**
	False alarms	15	14	13	10	11	**10**
	Chip amounts	126	128	126	123	124	124
	precision	0.8809	0.8906	0.8968	0.9186	0.9112	**0.9193**
	recall	0.9237	0.9406	0.9406	0.9406	0.9576	**0.9661**
	F1-score	0.9017	0.9149	0.9181	0.9294	0.9338	**0.9421**
	FoM	0.8195	0.8409	0.8473	0.8672	0.8759	**0.8906**

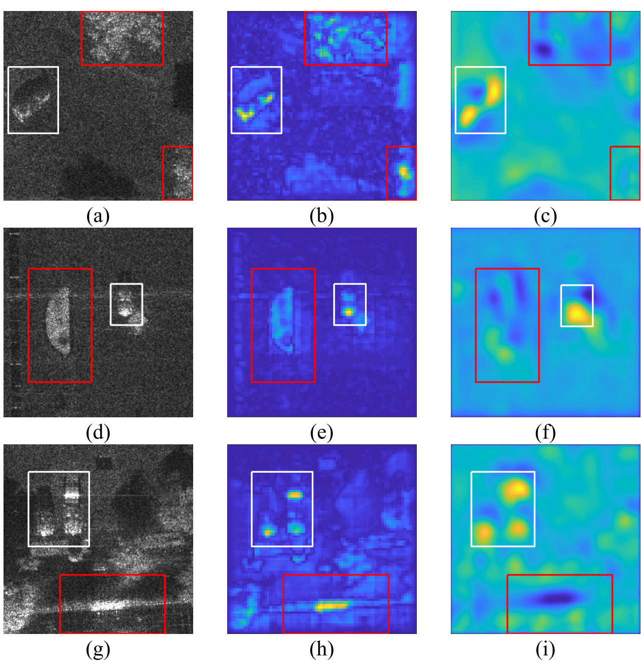


Fig. 8. Original SAR sub-images and their corresponding saliency maps obtained by the modified Itti's method and the proposed method separately, where white rectangles represent targets and red rectangles represent some clutter. (a), (d), and (g) Original SAR sub-images. (b), (e), and (h) Corresponding saliency maps via the modified Itti's method. (c), (f), and (i) Corresponding saliency maps via the proposed method.

performance has an accuracy higher by almost 15% $F1$-score and 13% FoM.

TABLE V
COMPLEXITIES OF THE S-SSD AND OTHER DEEP LEARNING-BASED DETECTION METHODS

	parameters	speed (FPS)
Faster R-CNN	134.7M	13
SSD	26.3M	82
S-SSD	45.0M	50

IV. CONCLUSION

In this article, we propose a unified end-to-end detection network, S-SSD, in which we integrate the saliency information into a deep learning-based detection network. By end-to-end learning, on the one hand, the saliency information can be further learned under supervision to generate a better saliency map and, on the other hand, the fine-tuned saliency information can guide the network to emphasize informative regions, making the representation capability improved. Furthermore, the dense connection structure, which utilizes multiscale features for prediction in the proposed S-SSD method can introduce more context information and mine the correlated complementary advantages across scales, which brings a significant performance improvement with fewer parameters. The experimental results based on the measured SAR images show that the proposed S-SSD target detection method has a better performance than the conventional CFAR methods and other deep learning-based detection methods, which demonstrates the effectiveness of the proposed method.

Although the proposed method has a significant improvement in performance, the speed of the proposed method is a little slower than the original SSD. In the future, we will explore the means of parallel computing and algorithm optimization to increase the speed.

REFERENCES

[1] L. M. Novak, M. C. Burl, and W. W. Irving, "Optimal polarimetric processing for enhanced target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 1, pp. 234–244, Jan. 1993.

[2] C. Alberola-Lopez, J. R. Casar-Corredera, and G. D. Miguel-Vela, "Object CFAR detection in gamma-distributed textured-background images," *IEE Proc.—Vis., Image Signal Process.*, vol. 146, no. 3, pp. 130–136, Jun. 1999.

[3] H. Dai, L. Du, Y. Wang, and Z. Wang, "A modified CFAR algorithm based on object proposals for ship target detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1925–1929, Dec. 2016.

[4] Z. Wang, L. Du, and H. Su, "Target detection via Bayesian-morphological saliency in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5455–5466, Oct. 2017.

[5] T. Li and L. Du, "Target discrimination for SAR ATR based on scattering center feature and K-center one-class classification," *IEEE Sensors J.*, vol. 18, no. 6, pp. 2453–2461, Mar. 2018.

[6] Y. Wang, L. Du, and H. Dai, "Target discrimination method for SAR images based on semisupervised co-training," *J. Appl. Remote Sens.*, vol. 12, no. 1, 2018, Art. no. 015004.

[7] Y. Sun, L. Du, Y. Wang, Y. Wang, and J. Hu, "SAR automatic target recognition based on dictionary learning and joint dynamic sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1777–1781, Dec. 2016.

[8] S. Deng, L. Du, C. Li, J. Ding, and H. Liu, "SAR automatic target recognition based on Euclidean distance restricted autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 7, pp. 3323–3333, Jul. 2017.

[9] L. Li, L. Du, and Z. Wang, "Target detection based on dual-domain sparse reconstruction saliency in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4230–4243, Nov. 2018.

[10] R. Min, H. Lan, Z. Cao, and Z. Cui, "A gradually distilled CNN for SAR target recognition," *IEEE Access*, vol. 7, pp. 42190–42200, 2019.

[11] Y. Kwak, W.-J. Song, and S.-E. Kim, "Speckle-noise-invariant convolutional neural network for SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 549–553, Apr. 2019.

[12] Z. Cui, C. Tang, Z. Cao, and S. Dang, "SAR unlabeled target recognition based on updating CNN with assistant decision," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1585–1589, Oct. 2018.

[13] G. Costante, T. A. Ciarfuglia, and F. Biondi, "Towards monocular digital elevation model (DEM) estimation by convolutional neural networks - application on synthetic aperture radar images," in *Proc. 12th Eur. Conf. Synth. Aperture Radar (EUSAR)*, Jun. 2018, pp. 1–6.

[14] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.

[15] R. Girshick, "Fast R-CNN," Apr. 2015, *arXiv:1504.08083*. [Online]. Available: https://arxiv.org/abs/1504.08083

[16] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," Jun. 2015, *arXiv:1506.01497*. [Online]. Available: https://arxiv.org/abs/1506.01497

[17] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," Jun. 2015, *arXiv:1506.02640*. [Online]. Available: https://arxiv.org/abs/1506.02640

[18] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 21–37.

[19] Z. Wang, L. Du, J. Mao, B. Liu, and D. Yang, "SAR target detection based on SSD with data augmentation and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 150–154, Jan. 2018.

[20] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 11, pp. 1254–1259, Nov. 1998.

[21] S. Liu, Z. Cao, and J. Li, "A SVD-based visual attention detection algorithm of SAR image," in *Proc. 2nd Int. Conf. Commun., Signal Process., Syst.* Cham, Switzerland: Springer, 2014, pp. 479–486.

[22] Z. Wang *et al.*, "Visual attention-based target detection and discrimination for high-resolution SAR images in complex scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1855–1872, Apr. 2018.

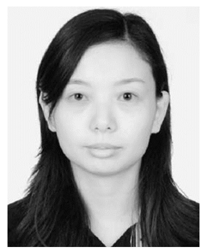
[23] T. Kan, Z. Yueting, C. Hongzhen, X. Wenchang, and W. Hongqi, "Radar image simulation of complex targets," in *Proc. IEEE CIE Int. Conf. Radar*, Oct. 2011, pp. 1319–1322.

[24] A. M. Law and W. D. Kelton, *Simulation Modeling and Analysis*. New York, NY, USA: McGraw-Hill, 2009.

[25] *SANDIA Mini SAR Complex Imagery*. Accessed: 2019. [Online]. Available: http://www.sandia.gov/radar/complex-data/index.html

[26] L. Mou, M. Schmitt, Y. Wang, and X. X. Zhu, "A CNN for the identification of corresponding patches in SAR and optical imagery of urban scenes," in *Proc. Joint Urban Remote Sens. Event*, Mar. 2017, pp. 1–4.

[27] J. Hu, L. Mou, A. Schmitt, and X. X. Zhu, "FusioNet: A two-stream convolutional neural network for urban scene classification using PolSAR and hyperspectral data," in *Proc. Joint Urban Remote Sens. Event*, Mar. 2017, pp. 1–4.



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